

The Tetrahedral Model of Learning:

Its origin, development and application

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Running Head: TETRAHEDRAL MODEL OF LEARNING

Very Good
Certainly relates
to the texts we used.
Would you be available
in the Fall to present
this to Ed 540? My
section will meet
on M + W from
10-11:50.
DLS

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In 1974, James Jenkins, an experimental psychologist with more than 20 years of experience, related to his colleagues that for most of those years he had been incorrectly interpreting the data which he gathered through his research (Jenkins, 1974). Jenkins had conducted research in verbal learning and paired associate learning from 1952 to 1969 all in pursuit of a better understanding of human memory. He and a colleague, Palermo had even produced a volume which documented the strength of one word's association with other words (Palermo & Jenkins, 1964). This volume became a standard resource for designing stringently controlled experiments in verbal learning. With this and other such resources, Jenkins conducted experiment after experiment and gathered more and more data for the understanding of the association model of human memory. Or so he thought, says Jenkins.

In 1969, Jenkins and a colleague, Thomas Hyde, decided that the data from as far back as 1952 (Jenkins & Russell, 1952) needed a review and perhaps a re-interpretation. What they soon realized was that all the data could be better explained in light of what Jenkins calls contextualism i.e. that experience occurs within a context with which it inextricably interacts. With subsequent experimentation (e.g., Hyde & Jenkins, 1973), Jenkins found support for his

idea that contextualism accounted for his previous data as well as some previously unexplained findings. In essence, what Jenkins had begun to do in 1969 was to review his data through another paradigm. He abandoned associationism - a reductionistic, idealistic view of memory - and adopted pragmatism which views events as having a "quality of the whole", a contextual quality.

As a result of this new paradigm, which emphasizes the context and the dynamics of the contextual variables, Jenkins propose a "model for memory experiments" (Jenkins, 1979). Originally he called the model the "Problem Pyramid" or the "Theorist's Tetrahedron." The model represents the variables found in any memory experiments and the interactions which can occur, particularly emphasizing the interactions that occur. That is, Jenkins proposed that basically four classes of variables exist: Orienting tasks, material characteristics, subject characteristics, and criterial characteristics. Orienting tasks are those types of behaviors a subject employs to learn the material e.g. verbally rehearsing a word or number to be remembered. Materials characteristics refer to the different types of stimulus materials which are presented to be learned e.g., pictures or words. Subjects characteristics vary as individual differences do e.g., young vs. old, readers vs. non-readers. Criterial characteristics are the different types of tests used to measure performance e.g., recall, or

cued recall. These four variables form the four vertices of the tetrahedron (see Figure 1). The lines between the vertices represent the possible interaction between the variables.

Insert Figure 1 about here

A typical memory experiment will hold two of the variable types constant and manipulate the other two. With the tetrahedral model, Jenkins was able to categorize most memory experiments onto one of the six lines of interaction between two points: 1. orienting task - criterial task interaction; 2. orienting task - materials interaction; 3. orienting task - subject interaction; 4. criterial task - materials interaction; 5. criterial task - subject interaction; and 6. materials - subject interaction.

Bransford (1979) applied Jenkins tetrahedral model to the learning process in general. He recognized that the tetrahedron provided a workable model for understanding the complexities of human cognition not just memory experiments. Others (Brown, Campione, & Day, 1981 and Snowman, 1986) have gone one step farther and applied the model to academic studying: In an academic situation, the orienting tasks are the tactics or study skills a student can use to learn the material. The materials characteristics are those of what

is to be learned e.g. textbook, math, lists, abstract, motor skill. The subject, or more properly, the learner's characteristics would include such factors as experience, motivation, handicaps, gifts, and preferences. The criterial characteristics describe the type of test used in the classroom e.g., essay, multiple choice, written report, oral presentation. Clearly, the application of the tetrahedral model is true to the original assumption of contextualism. The tetrahedral model describes well the varied and dynamic context in which academic learning takes place. Unfortunately for the student, she cannot hold two of the variables constant as you do in laboratory settings and have only two variables with which to be concerned.

The value of the tetrahedral model in academic learning lies not only with its ability to describe the situation but with its emphasis upon the interactions among the variables. To study appropriately and well the student must understand the particular context of his situation. The student will ask herself these types of questions: "What material is to be learned?" "What are its characteristics?" "How do I, the student, deal with this type of material?" "What types of study activities can I use?" "What type of test will be given?" "Will this affect how I study or what tactics I use?" "Do I want to study this?" "How much time do I have?" The model demonstrates its heuristic character by providing the means to answer these questions. The model

with its four points allows the student to consider all aspects of the learning situation.

At this point the tetrahedral model becomes useful in strategic thinking. Awareness and knowledge of the whole quality of the learning event is crucial to strategic thinking about the situation. Knowledge about your own knowledge and cognitive processes are important to your ability to think strategically. This type of knowledge and "thinking about thinking" is known as metacognition (Anderson, 1979; Brown, 1980; Thomas & Rohwer, 1986). Strategic behavior - analysis and planning, implementation, monitoring, and modification of a plan - depends upon metacognitive factors. Some of the beauty of the tetrahedral model is that it describes succinctly the types of data needs to be a successful strategic thinker. The four points of the student's tetrahedron ("student's t!") are seemingly common-sense. Indeed, one judgment of a model's utility is to what extent you slap your forehead and say "Yes, of course!" after seeing the model. The tetrahedral model elicits just that type of response. As Jenkins mentioned the four points represent most memory experiments. They also represent necessary metacognitive factors. For instance, the four points of the model coincide with the four metacognitive knowledge variables needed for learning mentioned in an article by Armbruster, Echols and Brown (1982). These four variables are text,

task, strategies (activities), and learner characteristics. Thomas and Rohwer (1986) mention similar variables in an article reviewing strategic behavior in academic settings in which they state "studying is context dependent" (p. 21): criterion relevance, clarity of criterion, processing load, availability of text aides, enhancement tactics, and the student's prior experience. Snowman (1986) nicely describes the relationship of the tetrahedral model to strategic thinking: "Strategic skills involve effectively analyzing a learning situation using the tetrahedral model as a guide and formulating a learning plan" (p. 245).

The tetrahedral model, then, has the potential to teach the student about the contextual aspects of the learning task i.e. studying. Furthermore, the model can demonstrate the use of metacognitive knowledge in strategic thinking. It can be used to introduce and name the components of strategic behavior to the student who, probably unknowingly, already has some familiarity with them. Finally, with the two purposes mentioned above fulfilled, the student may learn direct application of metacognition and strategic behavior to academic tasks. The student, perhaps for the first time, has concrete concept to the all too often vague, shadowy, and elusive meaning of "studying."

The teacher in the classroom, too, might benefit from application of the tetrahedral model. For example, considering the importance of "clarity of criterion" for

studying, the teacher can be sure to provide adequate information about the criterion i.e., the test and the objectives for the lessons. The teacher can also provide example tactics for studying particular types of material or help a student to be aware of his own preferences or strengths and weakness regarding the use of different study tactics. The teacher will be teaching, to a certain extent, strategic behavior. Providing adequate information and the resources to gain the necessary metacognitive knowledge with examples of how to use the information will help the student in his studying tasks and in learning.

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Tetrahedral Model

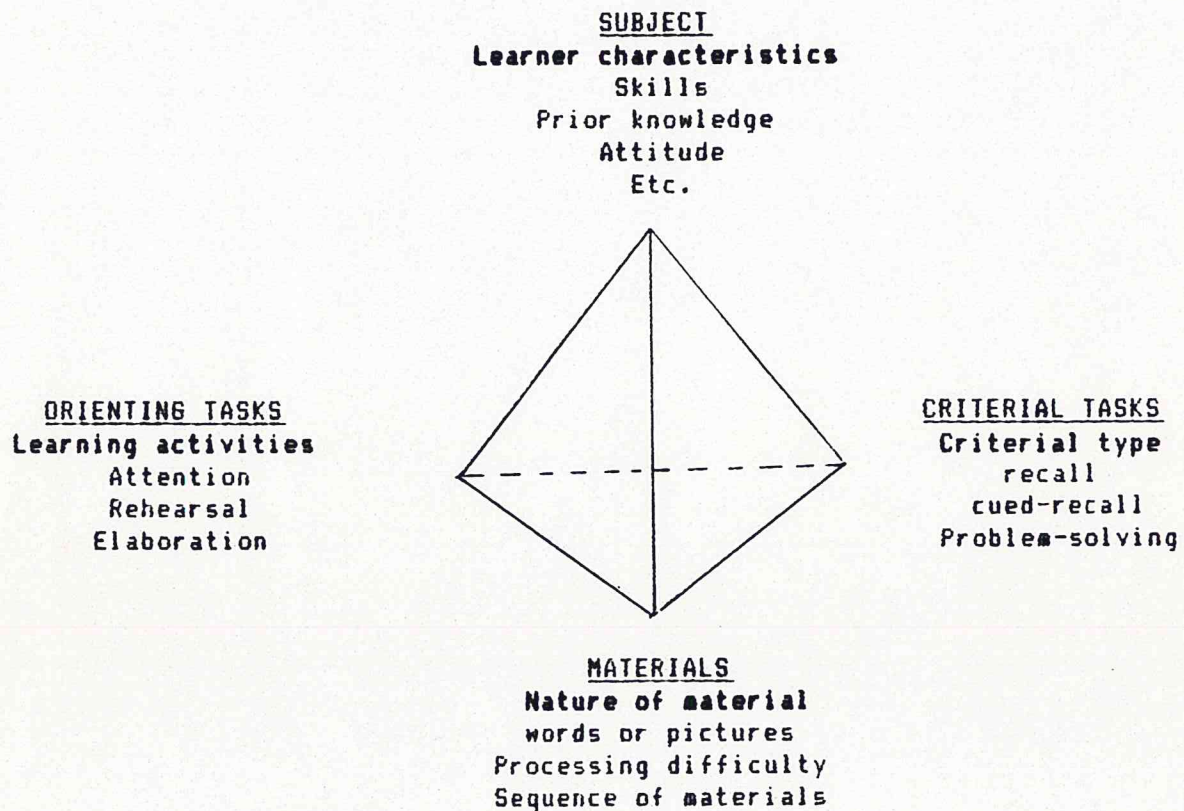


Figure 1: The tetrahedral model of learning. (Adapted from Bransford, 1979. Used without permission.)

Tetrahedral Model

